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(54) **ADJUSTABLE MEDICAL DEVICE**

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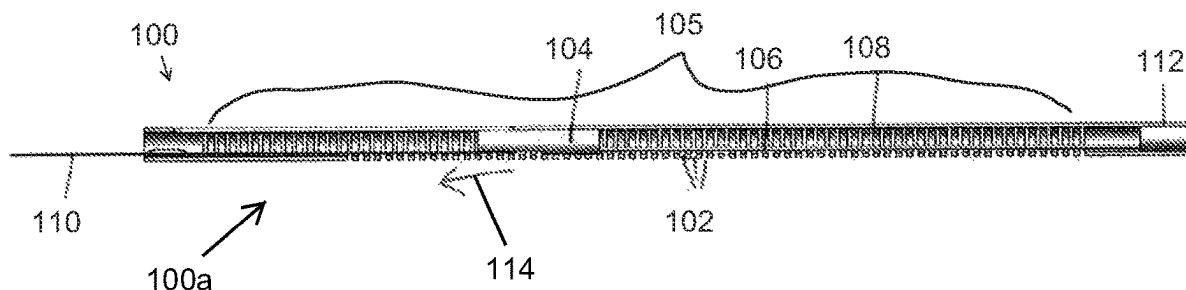
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(57) **ABSTRACT**

Medical devices that can be delivered in a first, flexible
configuration and subsequently locked into a second, rigid
configuration are provided. The device comprises an elon-
gate body having a plurality of slots openings along a
bending portion of the body. Using a tension member to
shorten the distance between two points along the elongate
body causes the device to curve. The device can be locked
into the curved configuration.

22 Claims, 10 Drawing Sheets



(58) **Field of Classification Search**

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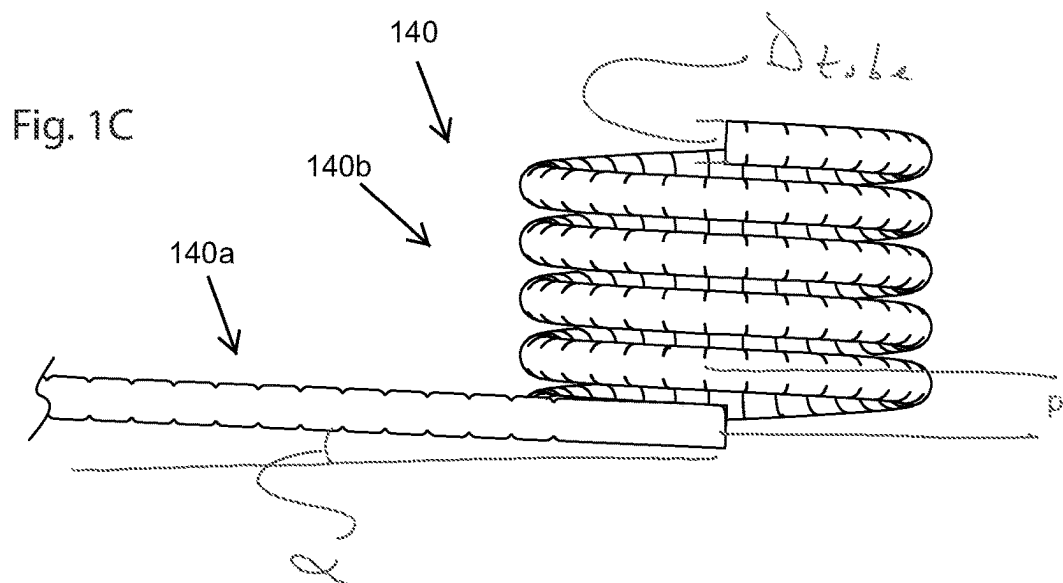
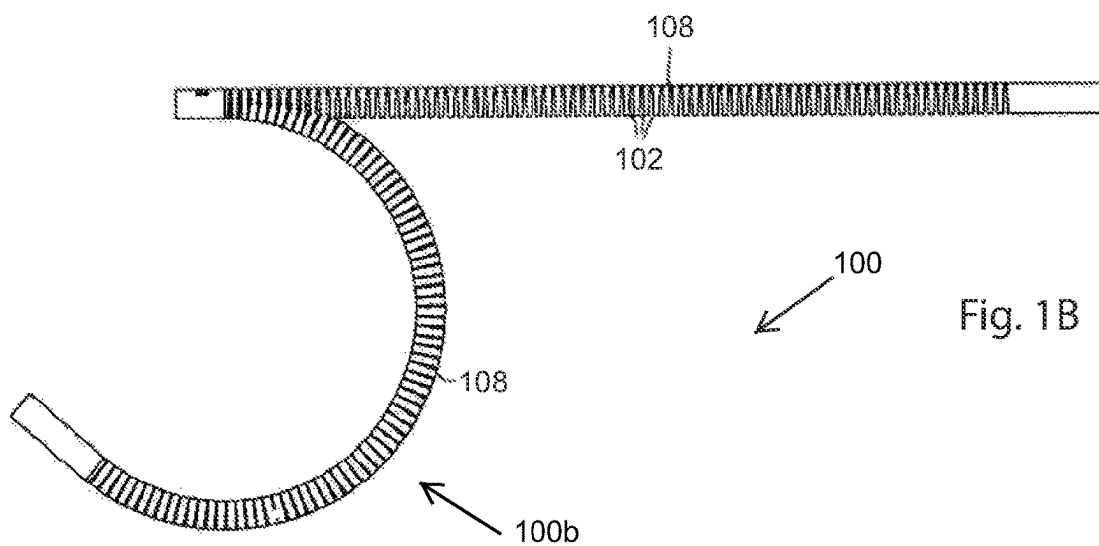
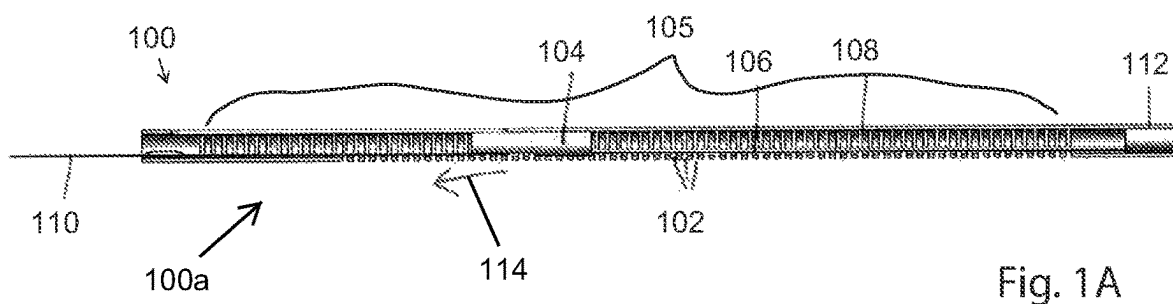
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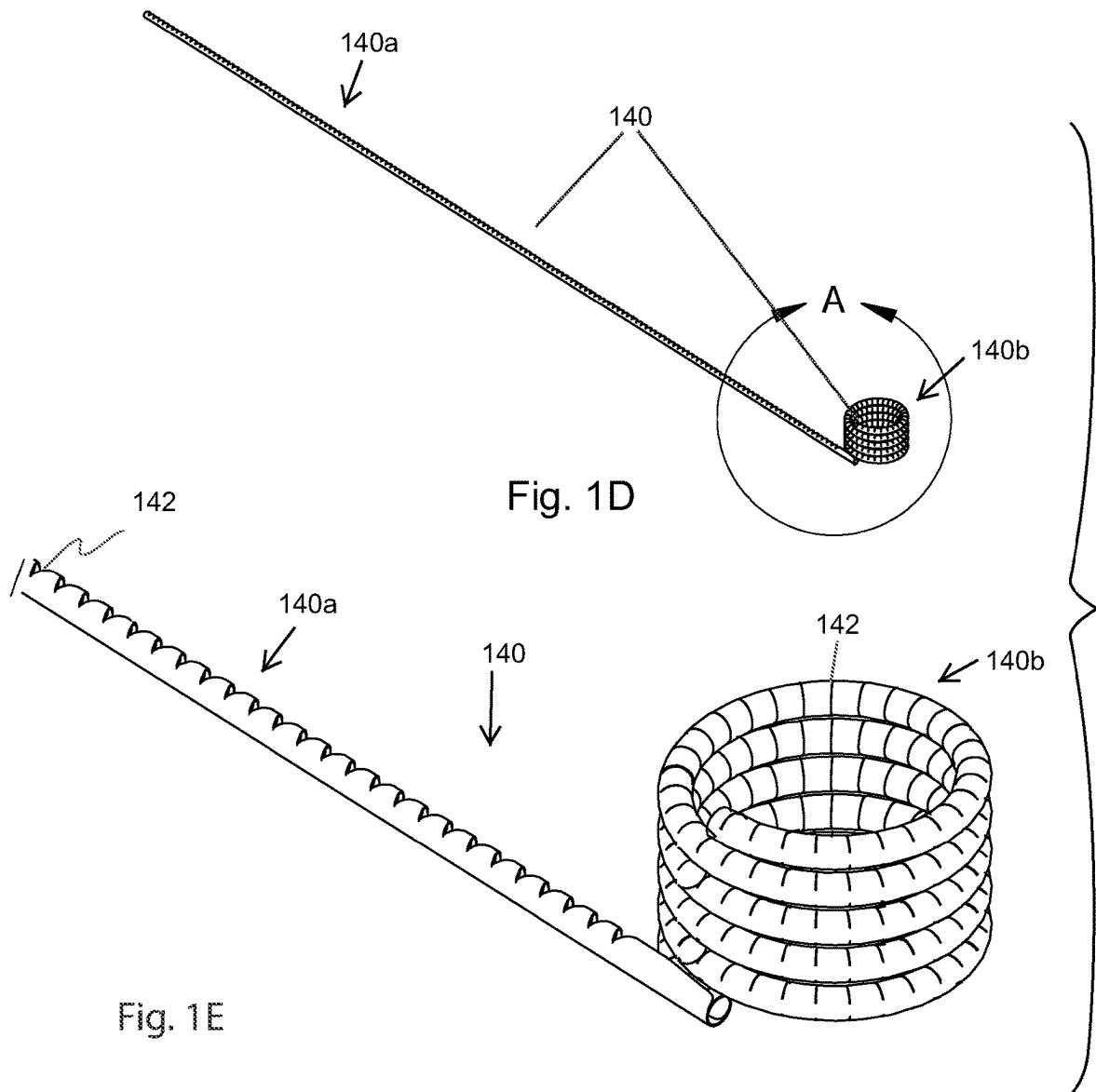
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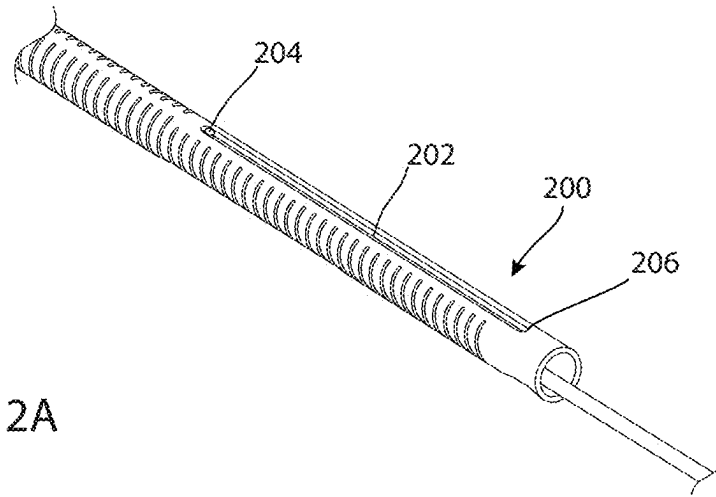


Fig. 2A

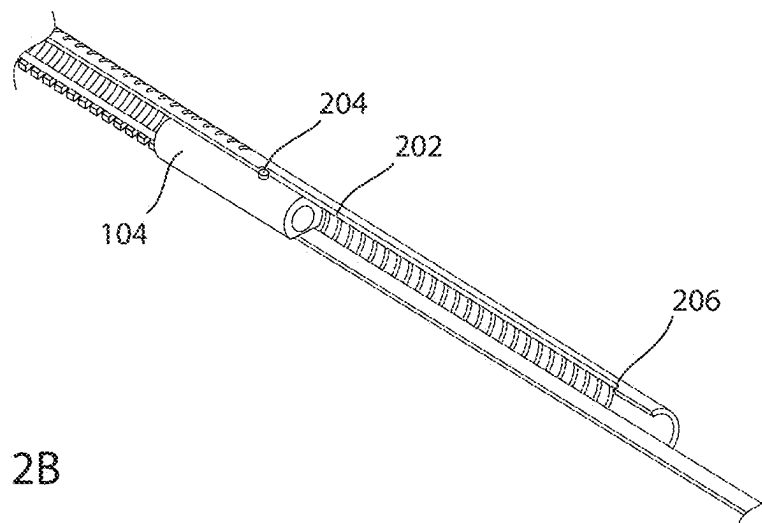
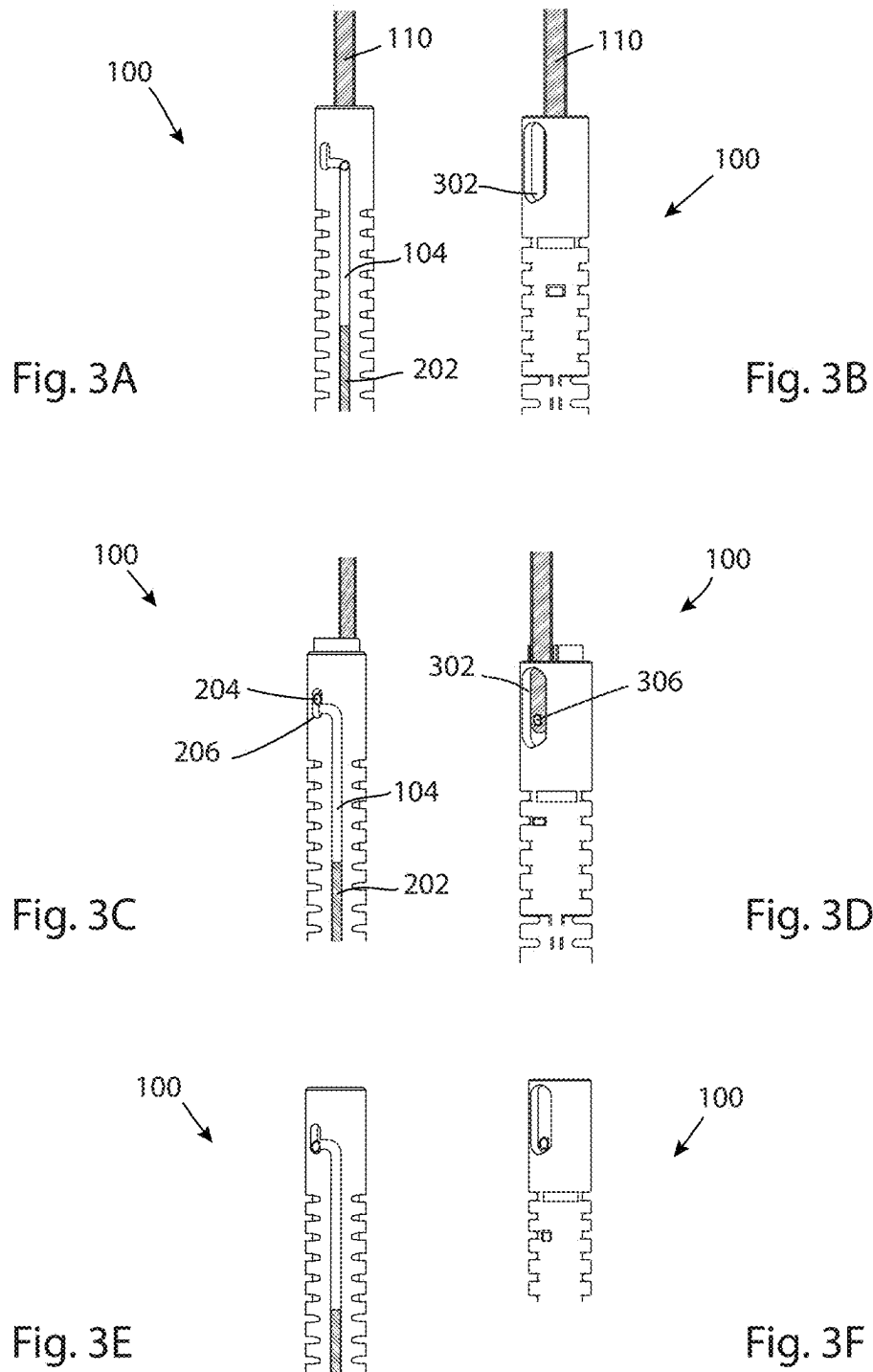


Fig. 2B



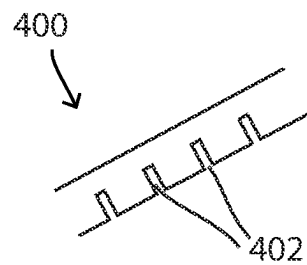


Fig. 4A

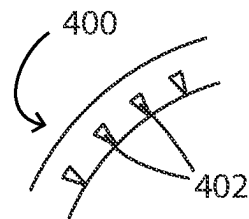


Fig. 4B

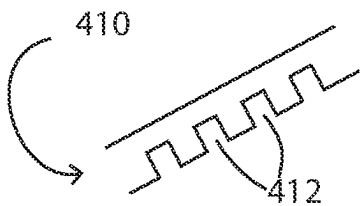


Fig. 4C

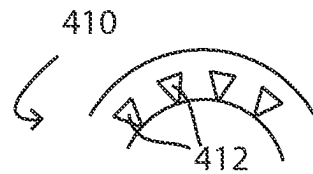


FIG. 4D

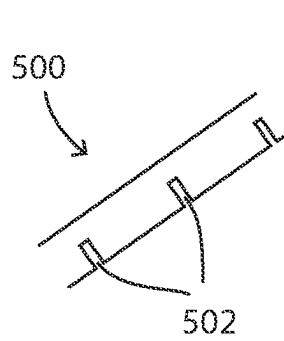


Fig. 5A

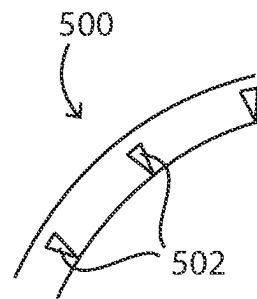


Fig. 5B

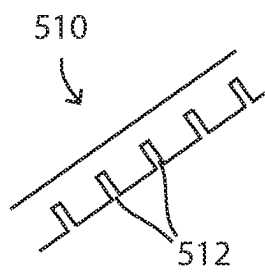


Fig. 5C

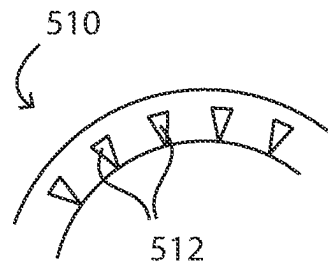


Fig. 5D

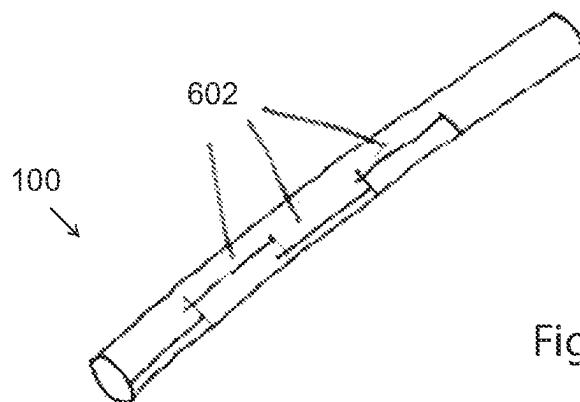


Fig. 6

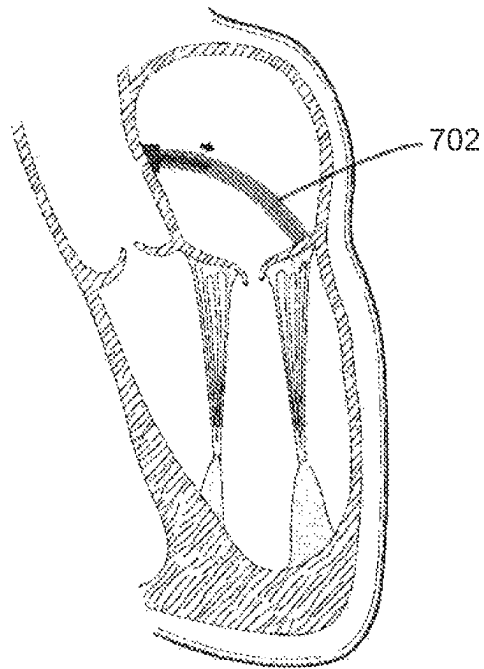


FIG. 7A

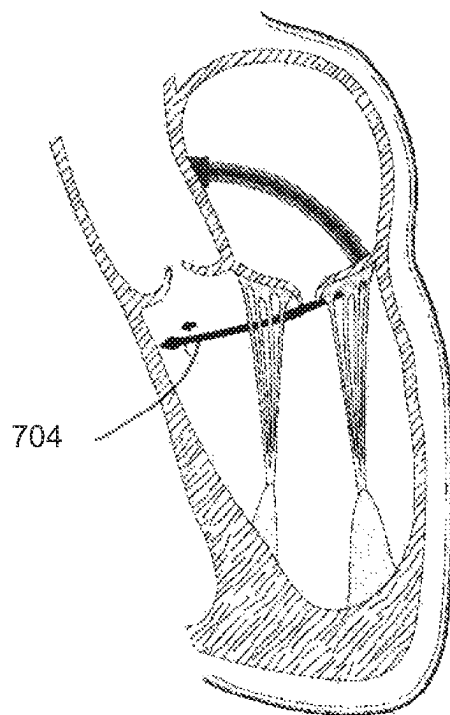


FIG. 7B

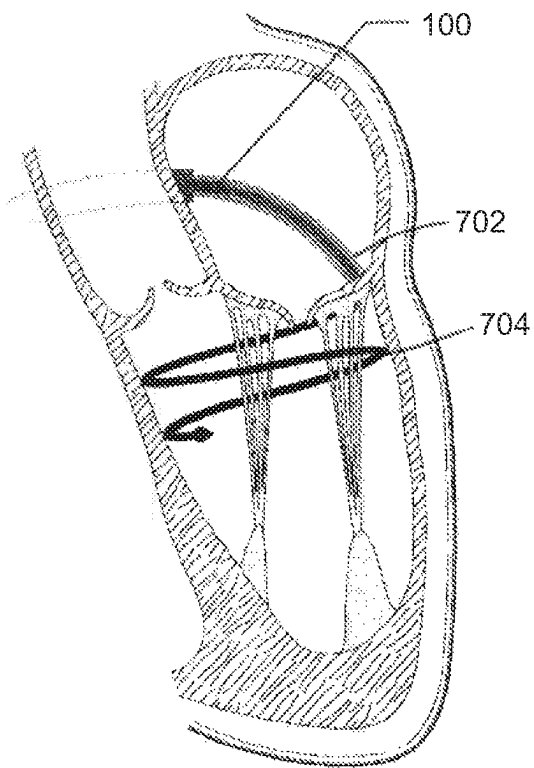


FIG. 7C

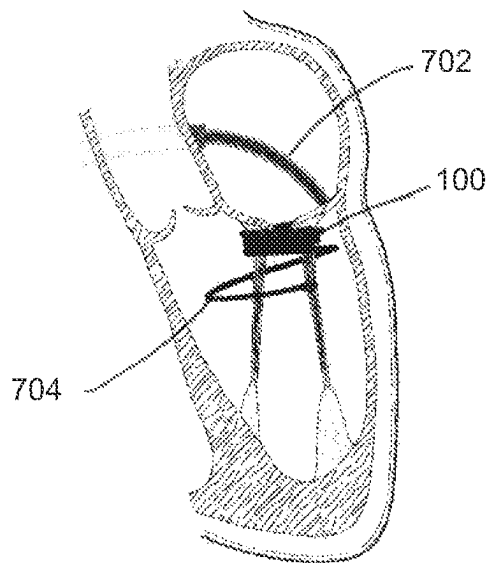


FIG. 7D

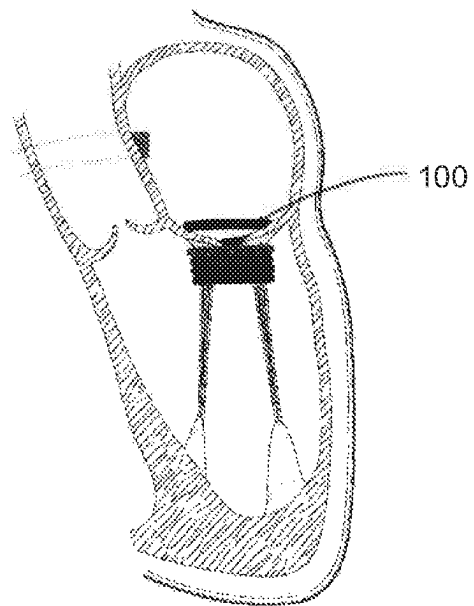


FIG. 7E

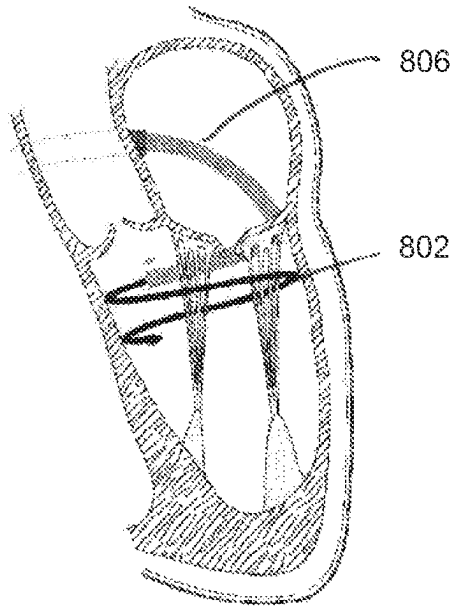


FIG. 8A

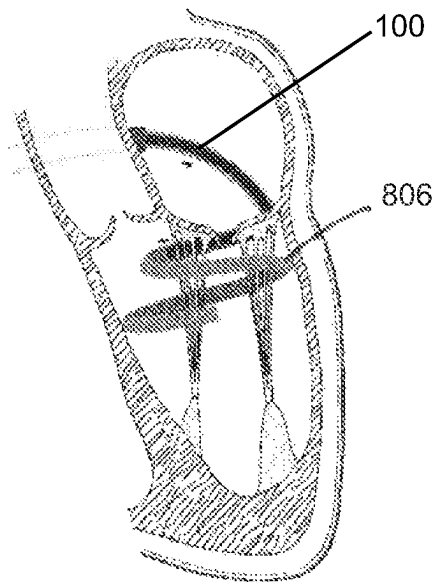


FIG. 8B

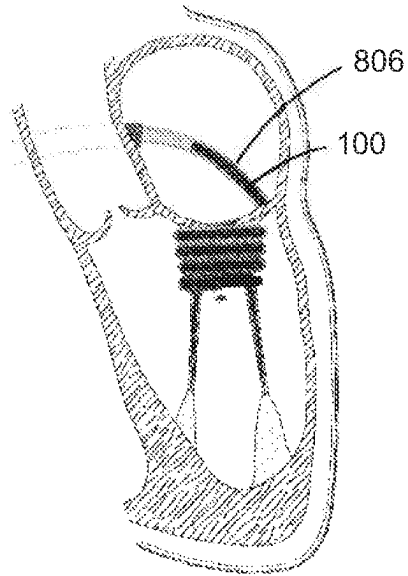


FIG. 8C

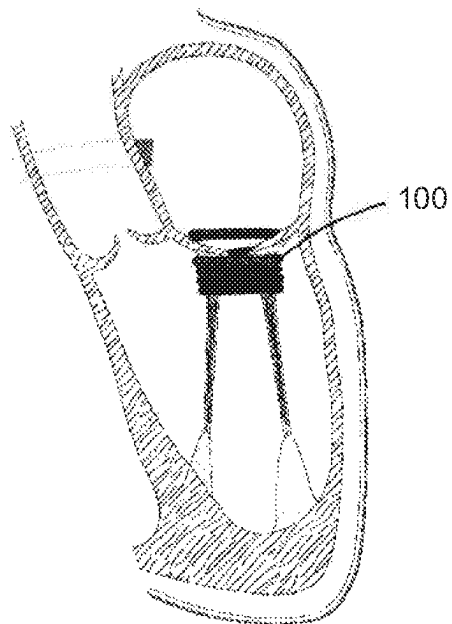


FIG. 8D

ADJUSTABLE MEDICAL DEVICE**CROSS-REFERENCE**

This PCT application claims the benefit of U.S. Provisional Application No. 62/748,162, filed Oct. 19, 2018, entitled "Adjustable Medical Device"; which is incorporated herein by reference for all purposes in its entirety.

The subject matter of this application is related to that of U.S. Provisional Application No. 62/720,853, filed Aug. 21, 2018, entitled "Prosthetic Cardiac Valve Devices, Systems, and Methods"; U.S. patent application Ser. No. 16/546,901, filed Aug. 21, 2019, entitled "Prosthetic Cardiac Valve Devices, Systems, and Methods"; U.S. Provisional Application No. 62/742,043, filed Oct. 5, 2018, entitled "Prosthetic Cardiac Valve Devices, Systems, and Methods"; U.S. Provisional Application No. 62/755,996, filed Nov. 5, 2018, entitled "Prosthetic Cardiac Valve Devices, Systems, and Methods"; U.S. patent application Ser. No. 16/594,946, filed Oct. 7, 2019, entitled "Prosthetic Cardiac Valve Devices, Systems, and Methods"; U.S. Provisional Application No. 62/784,280, filed Dec. 21, 2018, entitled "Prosthetic Cardiac Valve Devices, Systems, and Methods"; U.S. Provisional Application No. 62/813,963, filed Mar. 5, 2019, entitled "Prosthetic Cardiac Valve Devices, Systems, and Methods"; U.S. Provisional Application No. 62/815,791, filed Mar. 8, 2019, entitled "Prosthetic Cardiac Valve Devices, Systems, and Methods"; U.S. Provisional Application No. 62/820,570, filed Mar. 19, 2019, entitled "Prosthetic Cardiac Valve Delivery Devices, Systems, and Methods"; U.S. Provisional Application No. 62/828,835, filed Apr. 3, 2019, entitled "Prosthetic Cardiac Valve Devices, Systems, and Methods"; U.S. Provisional Application No. 62/833,425, filed Apr. 12, 2019, entitled "Minimal Frame Prosthetic Cardiac Valve Delivery Devices, Systems, and Methods"; U.S. Provisional Application No. 62/833,430, filed Apr. 12, 2019, entitled "Prosthetic Cardiac Valve Delivery Devices, Systems, and Methods"; U.S. Provisional Application No. 62/851,245, filed May 22, 2019, entitled "Prosthetic Cardiac Valve Devices, Systems, and Methods"; U.S. Provisional Application No. 62/872,016, filed Jul. 9, 2019, entitled "Prosthetic Cardiac Valve Delivery Devices, Systems, and Methods"; U.S. Provisional Application No. 62/873,454, filed Jul. 12, 2019, entitled "Systems, Methods, and Devices for Expandable Sensors"; U.S. Provisional Application No. 62/879,979, filed Jul. 29, 2019, entitled "Prosthetic Cardiac Valve Devices, Systems, and Methods"; and U.S. Provisional Application No. 62/894,565, filed Aug. 30, 2019, entitled "Prosthetic Cardiac Valve Devices, Systems, and Methods"; which are incorporated herein by reference for all purposes in their entireties.

BACKGROUND

Shape-memory devices are commonly used for minimally-invasive medical procedures due to their ability to return to a desired configuration upon release from a delivery device (in which the shape-memory devices are typically constrained in a narrower profile delivery configuration). However, many shape-memory devices require specialized delivery devices and methods to maintain the devices in a delivery configuration which enables minimally-invasive delivery to a target tissue, which may be less than ideal. Additionally, shape-memory devices may not be sufficiently

strong enough to be applied in certain medical procedures which may otherwise be accessible for minimally-invasive techniques.

SUMMARY

It would therefore be desirable to provide a medical device which can be transitioned from a small delivery configuration to a larger delivered configuration minimally requiring specialized delivery devices or methods and/or which may be stronger than some commonly-used shape-memory devices. Not necessarily all such aspects or advantages are achieved by any particular embodiment. Thus, various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages taught herein without necessarily achieving other aspects or advantages as may also be taught or suggested herein.

The present disclosure generally relates to medical devices and more particularly to elongate medical devices capable of being transitioned from a smaller profile delivery configuration to larger profile delivered configuration.

In a first aspect, a medical device is provided. The device comprises an elongate body having a first configuration and a second configuration, the second configuration having a different shape from the first configuration; shape changing features on the elongate body configured to allow the elongate body to change shape from the first configuration to the second configuration; and a locking mechanism configured to lock the elongate body in the second configuration.

In another aspect, a medical device is provided. The device comprises an elongate body; a spine portion extending along a sidewall of the elongate body; a plurality of openings extending from the spine portion; a shuttle axially movable along the elongate body; a tension member connecting a connection point positioned distally of the shuttle and the shuttle; a pull member extending proximally from the shuttle; and a locking mechanism configured to hold the shuttle in a proximally pulled position, wherein the medical device comprises a relaxed configuration before the shuttle is pulled proximally and a stiffened configuration after the shuttle is pulled proximally, the stiffened configuration causing the medical device to assume a predetermined, curved configuration, and wherein the plurality of openings is configured to have a first shape in the relaxed configuration and a second shape, different from the first shape in the stiffened configuration.

In some embodiments, a width of the openings changes between the first shape and the second shape. The radius of curvature of the predetermined, curved configuration can be determined by a width of the openings. In some embodiments, the plurality of openings comprises openings of varying size. The spine portion can curve around at least a portion of the elongate body. In some embodiments, the spine portion is arranged in a helical pattern around the elongate body. A pitch or coil angle α of the predetermined, curved configuration can be determined by a pitch or coil angle of the spine portion. Alternatively, the spine can be straight and the slots angled relative to the spine. The device can comprise a second connection point proximal to the connection point and a second wire connecting the second connection point to a second shuttle positioned proximally to the second connection point. In some embodiments, the device comprises a plurality of shuttles and a plurality of locking mechanisms. In some embodiments, the locking mechanism comprises a pin attached to the shuttle positioned in and configured to slide along a track extending along a length of the elongate body. The track can twist as

it extends proximally, forming a locking area for the pin. In some embodiments, the elongate body further comprises a release window configured to allow disengagement of the pull member from the shuttle. The device can be configured to expose the pull member to the release window when the pin is moved to the locking area. In some embodiments, the locking mechanism comprises a flexible screw positioned proximally to the shuttle, the flexible screw configured to be screwed to the shuttle or a component attached to the shuttle. The flexible screw can be configured to be screwed to a spring attached to the shuttle. In some embodiments, the flexible screw is configured to be screwed to a clip attached to the shuttle. The device can be configured to be delivered by a catheter. In some embodiments, a center portion of the device is configured to allow passage of a guidewire there-through. The device can comprise a catheter through which the medical device can be inserted. In some embodiments, the device is configured to be connected to an end of a catheter.

In another aspect, a method of deploying a medical device is provided. The method comprises advancing an elongate medical device to a target area in a patient's body, the medical device comprising a plurality of openings extending from a spine portion of a wall of the anchor, the medical device advanced in a first slack configuration; moving the medical device into a stiffened configuration by pulling proximally on a shuttle movable along the medical device, causing a tension member connecting the shuttle to a distal connection point to pull the distal connection point proximally; and locking the medical device in the stiff, activated configuration, the activated configuration comprising a predetermined curved shape.

In some embodiments, the method comprising activating the anchor comprises pulling proximally on a pull member attached to the shuttle while maintaining the position of the proximal end of the anchor. Locking the anchor can comprise moving a pin into a stop position along a track. In some embodiments, locking the anchor comprises screwing a screw to a component attached to the shuttle.

In some embodiments, the target area comprises a native valve in a heart in the patient's body. In some embodiments, the method further comprises advancing the elongate medical device from a first side of the native valve to a second side of the native valve.

In some embodiments, the method further comprises capturing one or more structures on the second side of the native valve with the elongate medical device in the first slack configuration prior to moving the elongate medical device into the stiffened configuration. In some embodiments, the native valve comprises a mitral valve, the first side comprises a left atrium, the second side comprises a left ventricle, and the one or more structures comprise one or more chordae tendineae or native leaflets.

In some embodiments, the method further comprises, after moving the medical device into the stiffened configuration, expanding at least a portion of an expandable valve prosthesis within at least a portion the elongate medical device adjacent the native valve, thereby anchoring the valve prosthesis to the native valve. Expanding the valve prosthesis can capture one or more structures of the native valve between the valve prosthesis and the elongate medical device. In some embodiments, the native valve comprises a mitral valve, the first side comprises a left atrium, the second side comprises a left ventricle, and the one or more structures comprise one or more chordae tendineae or native leaflets.

In another aspect, a method of advancing a medical device to a target area is provided. The method comprises advancing an elongate body towards the target area and out of a distal end of a delivery catheter, the elongate body comprising a first curved shape as the elongate body is advanced towards the target area; locking at least a portion of the elongate body into the first curved shape; moving the elongate body into a second curved shape at or near the target area, the second curved shape different from the first curved shape; and locking the elongate body into the second curved shape. For example, a distal portion of the device can be curved during navigation from a distal end of a delivery catheter to the target area. At or near the target area, a larger or different portion of the device can be locked into a desired final shape. In alternate embodiments, the elongate body is delivered out of the delivery catheter such that it follows the shape of the delivery catheter and then upon delivery to the target area the elongate body is locked into its delivery curved shape.

In another aspect, an anchor for use in transcatheter mitral valve repair is provided. The anchor comprises an elongate body; a spine portion along a sidewall of the elongate body; a plurality of openings extending from the spine portion; a connection point fixed to the elongate body; a shuttle positioned proximally to the connection point and axially movable along the elongate body; a wire connecting the connection point and the shuttle; a pull member extending proximally from the shuttle; and a locking mechanism for holding the shuttle in a proximally pulled position, the proximally pulled position causing the anchor to assume a predetermined, curved configuration.

In some embodiments, the spine portion is arranged to traverse a helical path around the elongate body. In other embodiments the slots are angled relative to the spine. The anchor can comprise a second connection point proximal to the connection point and a second wire connecting the second connection point to a second shuttle positioned proximally to the second connection point.

In another aspect, a method of performing transcatheter mitral valve repair is provided. The method comprises advancing a guidewire or guide catheter through an opening in the atrial septum, across a location near the anteromedial commissure; advancing an elongate anchor comprising a plurality of openings extending from a spine portion of a sidewall of the anchor over the guidewire such that the anchor forms at least one coil around the chordae and one coil in the left atrium, the anchor advanced in a first slack configuration; activating the anchor into a stiffened configuration having a predetermined curved shape by pulling proximally on a shuttle movable along the anchor, causing a wire connecting the shuttle to a distal connection point to pull the distal connection point proximally; and locking the anchor in the stiffened configuration. In some embodiments, the anchor is delivered around the chordae and/or the leaflets. In the stiffened configuration, the anchor can be tightened around the chordae and/or the leaflets.

In another aspect, a device for performing non-occlusive ablation is provided. The device comprises an elongate body; one or more electrodes positioned along the elongate body; a spine portion along a sidewall of the elongate body; a plurality of openings extending from the spine portion; a connection point fixed to the elongate body; a shuttle positioned proximally to the connection point and axially movable along the elongate body; a tension member connecting the connection point and the shuttle; a pull member extending proximally from the shuttle; and a locking mechanism for holding the shuttle in a proximally pulled position,

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the proximally pulled position causing the device to assume a predetermined, curved configuration.

In another aspect, a method for performing non occlusive ablation is provided. The method comprises advancing an elongate ablation device comprising a plurality of openings extending from a spine portion of a sidewall of the ablation device through the vasculature to a target site to be ablated, the ablation device advanced in a first slack configuration; moving the device into a stiffened configuration having a predetermined, curved shape by pulling proximally on a shuttle movable along the device, causing a wire connecting the shuttle to a distal connection point to pull the distal connection point proximally; and energizing one or more electrodes positioned along the device to ablate tissue near the device. In some embodiments, the spine portion is in the outer most position when the device is in a stiffened configuration. The spine portion can comprise electrodes. In some embodiments, the entire spine portion can comprise a metal and act as the electrode.

In some embodiments, the method comprises locking the device into a stiffened configuration. The method can comprise positioning the device against a tissue site to be ablated. In some embodiments, locking the ablation device causes the one or more electrodes to be in contact with the target site.

INCORPORATION BY REFERENCE

All publications and patent applications mentioned in this specification are incorporated herein by reference in their entirety to the same extent as if each individual publication or patent application was specifically and individually indicated to be incorporated by reference.

BRIEF DESCRIPTION OF THE DRAWINGS

The novel features of the disclosure are set forth with particularity in the claims that follow. A better understanding of the features and advantages of the present disclosure will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the present disclosure are utilized, and the accompanying drawings of which:

FIGS. 1A and 1B show side and top views, respectively, of an adjustable medical device having undeployed and deployed configurations, according to embodiments of the present disclosure.

FIGS. 1C, 1D, and 1E show side and perspective views of an adjustable medical device in a relaxed and in a stiffened configuration, according to embodiments of the present disclosure.

FIGS. 2A and 2B illustrate perspective and section views, respectively, of a locking mechanism on an adjustable medical device, according to embodiments of the present disclosure.

FIGS. 3A-3F show magnified side views of a locking and release mechanism on an adjustable medical device, according to embodiments of the present disclosure.

FIGS. 4A-5D show schematics of various embodiments of shape changing features for adjustable medical devices, according to embodiments of the present disclosure.

FIG. 6 illustrates a perspective view of another adjustable medical device, according to embodiments of the present disclosure.

FIGS. 7A-7E are section views of the left side of the heart illustrating a method of deploying an adjustable medical device at the mitral valve, according to embodiments of the present disclosure.

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FIGS. 8A-8D are section views of the left side of the heart illustrating another embodiment of a method of deploying a, adjustable medical device at the mitral valve, according to embodiments of the present disclosure.

DETAILED DESCRIPTION

In the following detailed description, reference is made to the accompanying figures, which form a part hereof. In the figures, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, figures, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the scope of the subject matter presented herein. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are explicitly contemplated herein.

Although certain embodiments and examples are disclosed below, inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses, and to modifications and equivalents thereof. Thus, the scope of the claims appended hereto is not limited by any of the particular embodiments described below. For example, in any method or process disclosed herein, the acts or operations of the method or process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various operations may be described as multiple discrete operations in turn, in a manner that may be helpful in understanding certain embodiments, however, the order of description should not be construed to imply that these operations are order dependent. Additionally, the structures, systems, and/or devices described herein may be embodied as integrated components or as separate components.

For purposes of comparing various embodiments, certain aspects and advantages of these embodiments are described. Not necessarily all such aspects or advantages are achieved by any particular embodiment. Thus, for example, various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may also be taught or suggested herein.

Provided herein are embodiments of an adjustable medical device. The device can comprise a slackened configuration and one or more stiffened configurations, the stiffened configurations comprising a predetermined curved shape. The device can be locked into the stiffened configuration(s). For example, in some embodiments, the device is percutaneously advanced to a target area within a patient in a slackened or relaxed configuration. Once the device is deployed to or positioned within a target area, it can be locked into a stiffened configuration comprising a predetermined curved shape. Such a device can be used in a number of different applications, as described below.

FIG. 1A depicts an embodiment of an adjustable medical device **100** in an undeployed, straightened configuration **100a**. The device **100** can comprise an elongate body. For example, in some embodiments, the device **100** comprises an elongate, tubular body. The device **100** can be shaped to be delivered percutaneously through or at the end of a catheter. The device **100** comprises a plurality of slots or openings **102** surrounding or extending from a spine portion **108**. The portion(s) of the device **100** comprising the openings **102** is the collapsing portion(s) **105** of the device **100**.

The spine portion **108** of the device may be defined by an area of the bending portion **105** of the device **100** without openings **102**. The collapsing of the collapsing portion **105** allows the device **100** to bend at the spine portion **108** of the device **100**. In some embodiments, the spine portion **108** can be thin, for example, comprising a width of about 0-10% of a circumference of the elongate body. In other embodiments, the spine portion **108** can be wider, for example, comprising greater than about 10% of a circumference of the elongate body. In some embodiments, the openings **102** surrounding the spine portion **108** are wedged-shaped. Other configurations for the openings are also possible (e.g., slit-shaped, rounded, etc.) as described herein. In some embodiments, the device **100** can be formed from a hypotube. The openings **102** can be formed by laser cutting, a blade, or the like as will be understood by one of ordinary skill in the art based on the description herein. In some embodiments, the device **100** may comprise a flexible covering or sleeve (e.g., a polymer covering). The cover can help protect surrounding tissue from trauma which may be caused by the openings **102** as the device **100** bends.

The shape or amount of bending of the device **100** in the stiffened configuration **100b** is defined by the difference in length between the outer edge of the curve (the spine portion **108**) and the inner edge of the curve comprising the openings **102**. This difference in length is determined by the width of the openings **102**, the sum of which defines the difference in length between the inner and outer edge of the curve.

The device **100** may comprise a plug **112** positioned at a distal portion of the device **100**. A tension member **106** (e.g., a wire, an elastic wire, or spring, or the like) may connect the plug **112** and a shuttle **104**. The tension member **106** can hold the length of the inner radial surface of the curved portion of the device **100** at its minimal length. It will be appreciated by one of ordinary skill in the art based on the description herein that, in some embodiments, the device **100** may not comprise a plug **112**, and the tension member **106** may be connected via another means to another distal point of the device **100**. The shuttle **104** is axially movable along a length of the device **100**. The shuttle **104** can comprise a cylinder, in some embodiments. Other configurations are also possible (e.g., spherical shape, elliptical shape, rectangular, etc.), as will be understood by one of ordinary skill in the art based on the description herein. The shuttle **104** can be movable within the device **100**, over the device **100**, or partially within and over the device **100**. A pull member **110** (e.g., wire, rod, etc.) may extend proximally from the shuttle **104**, and allow the shuttle **104** to be pulled proximally while a proximal end of the device **100** is being held or pushed distally, thereby shortening the distance between the shuttle **104** and the proximal end of the device **100**. Alternatively, the proximal end of the device **100** can be pushed distally while the shuttle **104** and/or pull member **110** is held in place. The reduction in distance between the shuttle **104** and the proximal end of the device **100** may define the minimum length of the inner radial surface of the device **100**.

The pull member **110** can be used to pull proximally, in the direction indicated by the arrow **114**, on the shuttle **104**. The shuttle **104** moving proximally may cause the shape of the openings **102** to change. The width of the openings **102** may decrease as the shuttle **104** moves proximally, thus shortening the side of device **100** positioned away from spine **108**. The shortening of one side of the device **100** causes the device **100** to curve away from the spine portion **108**, as shown in FIG. **1B**.

In the stiffened configuration **100b**, the device **100** can comprise a variety of shapes. For example, the device **100** shown in FIG. **1A** may deform into the shape of a portion of a circle as shown in FIG. **1B**. In some embodiments, the device **100** comprises a helical shaped device **140** as in FIGS. **1C** and **1D** in the stiffened configuration **100a**, **140a**. The helical shape can have a coil angle α , and a pitch p . The coil angle α and coil pitch p of the device **140** in the stiffened configuration **140b** can be determined by the angle and pitch of the spine portion **108** of the device **140**. The device **140** can comprise any combination of a spiral shape, a conical shape, a frustoconical shape, a helical shape, etc. As shown in the device **140** of FIGS. **1C** and **1D**, the openings **142** can be arranged at an angle relative to the spine **108** to facilitate helical coiling.

FIG. **1D** illustrates a device **140** in a relaxed, straight configuration **140a**, and in a stiffened, coiled configuration **140b**. FIG. **1E** illustrates a magnified view of the portion of FIG. **1D** shown in section A. The slots or openings **142** are shown open in the relaxed, straight configuration **140a**. In the stiffened, coiled configuration **140b**, the slots or openings **142** may be collapsed, allowing the device **100** to curve.

The device **100** can comprise a locking mechanism to lock the shuttle **104** in a proximally pulled position. FIGS. **2A** and **2B** show an embodiment of a locking mechanism **200**. FIG. **2A** shows an external view of the device **100**. FIG. **2B** shows a cut-away view of the device **100**, showing the shuttle **104** within the device **100**. The locking mechanism **200** may comprise a track **202**. A pin **204** may be attached to the shuttle **104** and configured to slide along the track **202**. The track **202** may lead to a slot or opening **206** having a distal wall against which the pin **204** can rest, thereby preventing the pin **204** from moving distally and locking the shuttle **104** into position. Other locking mechanisms **200** are also possible. For example, the shuttle **104** can be locked into position using a flexible screw running on the upper side of the shuttle **104**. The screw can be configured to be screwed into an anchor portion of the device **100** (e.g., plug **112**). Turning the screw can cause the shuttle **104** to move towards or away from the anchor. The screw can extend along the length of the spine portion **108**. The screw may not extend out of the anchor when in the stiffened configuration as the spine **108** maintains its length between the collapsed and stiff configurations. In other embodiments, a screw and clip mechanism can be used.

FIGS. **3A-3B** illustrate a detailed view of a locking mechanism **300** like the locking mechanism **200**, further comprising a disengagement feature configured to allow the pull member **110** to disengage from the shuttle **104** when the shuttle **104** is pulled into a locked position. FIGS. **3A** and **3B** illustrate the device **100** with the shuttle **104** pulled proximally along the track **202**, but not yet placed into a locked position. FIG. **3A** shows the track **202** and the locking opening **206**. FIG. **3B** shows a release window **302** positioned at a different portion of the device **100**. As the pin **204** is moved into the locking opening **206**, the shuttle **104** may twist, as shown in FIGS. **3C** and **3D**. The twisting of the shuttle **104** may position the connection between the shuttle **104** and the pull member **110** in the release window **302**, shown in FIG. **3D**. The connection may comprise an aperture **304** on the pull member **110** engaged with a pin **306** on the shuttle **104**. Moving the aperture **304** engaged with the pin **306** to the release window **302** may cause the pull member **110** to disengage from the shuttle **104**. FIGS. **3E** and **3F** show the device **100** after the pull member **110** has been disengaged and withdrawn.

Various factors can influence the radius of curvature of the device **100** in the locked or stiffened configuration **100b**. For example, the configuration of the spine **108** can influence the configuration of the locked curvature. A spine portion **108** that coils around the device **100** can produce a locked device with a helical configuration. The coil angle and pitch of the spine portion **108** can determine the shape of the stiffened device as described herein.

As noted above, the radius of curvature can be dependent on a total width of the openings. For example, FIGS. 4A and 4B show a device **400** with narrower openings **402** than those openings **412** shown in the device **410** of FIGS. 4C and 4D. The device **410** with the wider openings **412** has a deeper curve (smaller radius of curvature) (FIG. 4D) than that of the device **400** with the smaller openings **402** (FIG. 4B).

The density of the openings or the frequency of the openings along the length of the device can also determine a radius of curvature of the device, as a device with a higher density of openings has a higher total width of the openings. FIGS. 5A and 5B show a device **500** with fewer openings **502** than the device **510** of FIGS. 5C and 5D. The device **510** with a higher density of openings **512** has a higher radius of curvature (FIG. 5D) than that of the device **500** with the less dense openings **502** (FIG. 5B).

It will be understood by one of ordinary skill in the art that any of the devices described herein (e.g., **100**, **140**, **400**, **410**, **500**, **510**) may comprise any number, shape, or density of openings (e.g., **102**, **142**, **402**, **412**, **502**, **512**) as desired.

In some embodiments, the device **100** may comprise multiple locking mechanisms **602** along a length of the device **100** as shown in FIG. 6. In such a device **100**, multiple shuttles, tension members, and locking mechanisms can be used to separately and/or independently stiffen and lock various bending portions of the device **100**. Each bending portion may comprise a proximal anchor portion comprising a locking mechanism (e.g., similar to locking portion **206** of FIGS. 2A-2B) that can be placed into contact with the corresponding shuttle to cause the corresponding bending portion to curve. In some embodiments, such a configuration can be used to sequentially lock portions of the device **100** and/or to achieve more complex stiffened configurations. For example, as the device **100** is deployed from a delivery catheter, the deployed portions can be stiffened and locked. This sequential locking can be performed manually by a clinician or, alternatively, can be performed by an automated system (e.g., a delivery system). In some embodiments, the act of exposing the device **100** or the relative motion between the delivery catheter and the device **100** can cause the exposed portions to be curved and locked. In some embodiments, a portion of the device **100** can comprise a first shape for delivery and a second, different shape when deployed. For example, a tip of the device **100** can be curved during delivery to aid with navigation. The device **100** can be stiffened into a different shape when deployed. In some embodiments, multiple shuttles are used with the different locking mechanisms. In other embodiments, one shuttle can be locked into different positions, causing different portions of the device **100** to be curved. Allowing this sort of sequential stiffening of the device **100** can allow the device **100** to form the path it takes after exiting the delivery catheter, which can diminish the need for a guide wire or guide catheter.

In some embodiments, the device **100** can be used as a dock for anchoring a valve prosthesis during a mitral valve repair or replacement procedure, as shown in FIGS. 7A-7E. FIG. 7A illustrates delivery catheter **702** positioned through

the atrial septum after transseptal puncture. A guidewire **704** may be inserted into the heart via the delivery catheter **702** and through tissue at or near the anteromedial commissure, as shown in FIG. 7B. The guidewire **704** may be advanced such that it coils around the chordae tendineae in the left ventricle, as shown in FIGS. 7B and 7C. The device **100**, or any of the devices described herein, may then be advanced through the catheter **702** and over the guidewire **704**, as shown in FIG. 7D. The device **100** can be inserted in a slackened configuration **100a**. The shape of the device **100** may mimic the guidewire **704** during delivery to the chordae tendineae. Once the device **100** is deployed around the chordae tendineae, it can be locked into the stiffened configuration **100b**, as shown in FIG. 7E. As shown, the device **100** may form a helical shape in the stiffened configuration **100b** with at least one coil around the chordae tendineae in the left ventricle and one coil in the left atrium. In some embodiments, the entire device **100** may be located in the left ventricle, with no coils sitting in the left atrium after deployment. The guidewire **704** can be retracted once the device **100** is in place. In some embodiments, the device **100** may comprise multiple bending portions as described above with respect to FIG. 6. Retracting the guidewire **704** from the device **100** can be coordinated with locking the device **100** so that the device **100** is locked into shape from its distal end towards its proximal end as the distal tip of the guidewire **704** moves from the distal end towards the proximal end of the device **100** as it is retracted. A valve prosthesis (not shown) may be delivered to the mitral valve and anchored to the mitral valve with the device **100** after the device **100** is deployed as described in U.S. patent application Ser. Nos. 16/594,946 and 16/546,901, and U.S. Provisional Application Nos. 62/720,853, 62/742,043, 62/755,996, 62/784,280, 62/813,963, 62/815,791, 62/820,570, 62/828,835, 62/833,425, 62/833,430, 62/851,245, 62/872,016, 62/873,454, 62/879,979, and 62/894,565, previously incorporated herein by reference for all purposes.

FIGS. 8A-8D illustrate a similar method to that shown in FIGS. 7A-7E; however, in the method of FIGS. 8A-8D, the device **100** is deployed through a guide catheter **806** instead of over a guidewire **802**. FIG. 8A shows the delivery catheter **806** positioned through the atrial septum after transseptal puncture. A guidewire **802** may then be inserted through tissue at or near the anteromedial commissure and advanced such that it coils around the chordae tendineae in the left ventricle, as shown in FIG. 8A. The guide catheter **806** may be advanced over the guidewire **802**, as shown in FIG. 8B. The guidewire **802** may then be withdrawn. The device **100** may then be advanced through the guide catheter **806**, as shown in FIG. 8B. The guide catheter **806** may then be withdrawn, as shown in FIG. 8C. As shown in FIG. 7E, the device **100** may form a helical shape in the stiffened configuration with at least one coil around the chordae tendineae in the left ventricle and one coil in the left atrium. In some embodiments, the entire device **100** may be located in the left ventricle, with no coils sitting in the left atrium after deployment. The guide catheter **806** can be retracted once the device **100** is in place. In some embodiments, the device **100** may comprise multiple bending portions as described above with respect to FIG. 6. Retracting the guide catheter **806** from the device **100** can be coordinated with locking the device **100** so that the device **100** is locked into shape from its distal end towards its proximal end as the distal tip of the guide catheter **806** moves from the distal end towards the proximal end of the device **100** as it is retracted. A valve prosthesis (not shown) may be delivered to the mitral valve and anchored to the mitral valve with the device **100** after

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the device **100** is deployed as described in U.S. patent application Ser. Nos. 16/594,946 and 16/546,901, and U.S. Provisional Application Nos. 62/720,853, 62/742,043, 62/755,996, 62/784,280, 62/813,963, 62/815,791, 62/820, 570, 62/828,835, 62/833,425, 62/833,430 62/851,245, 62/872,016, 62/873,454, 62/879,979, and 62/894,565, previously incorporated herein by reference for all purposes.

It will be appreciated that the device **100** can be used in other configurations as well. For example, in some embodiments, the device **100** can be used for non-occlusive ablation. The device **100** can be advanced to an area to be treated in a slackened or relaxed configuration. Once the device **100** reaches the target area, the device **100** can be locked into a stiffened configuration. In some embodiments, the device **100** can be moved into the stiffened configuration without actually locking it into place. Instead, it can be held in the stiffened configuration during ablation, and then released to move the device **100** to another ablation site or to withdraw the device **100**. The stiffened configuration can comprise a predetermined curve configured to put the device **100** into contact with an area to be ablated. For example, the device **100** can be configured to coil around a perimeter of a blood vessel. The shape of the curve can be selected to ensure good contact between the device **100** and the area to be ablated. The spine portion **108** can be the outer most portion of the device **100** when in the stiffened configuration. The device **100** can comprise at least one electrode (e.g., 1, 2, 3, 4, 5, 6, 7, 8, or more electrodes) located such that they can be positioned against an area to be ablated in the stiffened configuration. In some embodiments, at least a portion of the device **100** itself (e.g., the spine portion **108**) can comprise a metal and serve as the electrode. In some embodiments, in which the device **100** is used for non-occlusive ablation, the device **100** comprises a lumen through which blood can flow.

In some embodiments, the device **100** can be used in an anchor for an embolic protection device such as a filter for use during a cardiac intervention procedure. The device **100** can advantageously be delivered in a small profile and then deployed to a larger desired shape. The devices described herein can have certain advantages over shape memory devices that are able to be delivered in a slack configuration and then assume a deployed shape upon delivery. The devices described herein can comprise greater stiffness and strength in the deployed position than shape memory devices in a deployed position. Shape memory devices having deployed configurations with a much greater diameter than the delivered configuration (e.g., a coil) must be delivered in a catheter strong enough to maintain the shape memory device in a straight configuration during delivery as the shape memory device will bias towards the larger diameter configuration. The devices of the current application are not biased during delivery and do not need to be held in place during delivery by a stronger or thicker catheter.

When a feature or element is herein referred to as being “on” another feature or element, it can be directly on the other feature or element or intervening features and/or elements may also be present. In contrast, when a feature or element is referred to as being “directly on” another feature or element, there are no intervening features or elements present. It will also be understood that, when a feature or element is referred to as being “connected”, “attached” or “coupled” to another feature or element, it can be directly connected, attached or coupled to the other feature or element or intervening features or elements may be present. In contrast, when a feature or element is referred to as being “directly connected”, “directly attached” or “directly

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coupled” to another feature or element, there are no intervening features or elements present. Although described or shown with respect to one embodiment, the features and elements so described or shown can apply to other embodiments. It will also be appreciated by those of skill in the art that references to a structure or feature that is disposed “adjacent” another feature may have portions that overlap or underlie the adjacent feature.

Terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. For example, as used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items and may be abbreviated as “/”.

Spatially relative terms, such as “under”, “below”, “lower”, “over”, “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if a device in the figures is inverted, elements described as “under” or “beneath” other elements or features would then be oriented “over” the other elements or features. Thus, the exemplary term “under” can encompass both an orientation of over and under. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly. Similarly, the terms “upwardly”, “downwardly”, “vertical”, “horizontal” and the like are used herein for the purpose of explanation only unless specifically indicated otherwise.

Although the terms “first” and “second” may be used herein to describe various features/elements (including steps), these features/elements should not be limited by these terms, unless the context indicates otherwise. These terms may be used to distinguish one feature/element from another feature/element. Thus, a first feature/element discussed below could be termed a second feature/element, and similarly, a second feature/element discussed below could be termed a first feature/element without departing from the teachings of the present disclosure.

Throughout this specification and the claims which follow, unless the context requires otherwise, the word “comprise”, and variations such as “comprises” and “comprising” means various components can be co jointly employed in the methods and articles (e.g., compositions and apparatuses including device and methods). For example, the term “comprising” will be understood to imply the inclusion of any stated elements or steps but not the exclusion of any other elements or steps.

As used herein in the specification and claims, including as used in the examples and unless otherwise expressly specified, all numbers may be read as if prefaced by the word “about” or “approximately,” even if the term does not expressly appear. The phrase “about” or “approximately” may be used when describing magnitude and/or position to indicate that the value and/or position described is within a reasonable expected range of values and/or positions. For

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example, a numeric value may have a value that is $\pm 0.1\%$ of the stated value (or range of values), $\pm 1\%$ of the stated value (or range of values), $\pm 2\%$ of the stated value (or range of values), $\pm 5\%$ of the stated value (or range of values), $\pm 10\%$ of the stated value (or range of values), etc. Any numerical values given herein should also be understood to include about or approximately that value, unless the context indicates otherwise. For example, if the value “10” is disclosed, then “about 10” is also disclosed. Any numerical range recited herein is intended to include all sub-ranges subsumed therein. It is also understood that when a value is disclosed that “less than or equal to” the value, “greater than or equal to the value” and possible ranges between values are also disclosed, as appropriately understood by the skilled artisan. For example, if the value “X” is disclosed the “less than or equal to X” as well as “greater than or equal to X” (e.g., where X is a numerical value) is also disclosed. It is also understood that the throughout the application, data is provided in a number of different formats, and that this data, represents endpoints and starting points, and ranges for any combination of the data points. For example, if a particular data point “10” and a particular data point “15” are disclosed, it is understood that greater than, greater than or equal to, less than, less than or equal to, and equal to 10 and 15 are considered disclosed as well as between 10 and 15. It is also understood that each unit between two particular units are also disclosed. For example, if 10 and 15 are disclosed, then 11, 12, 13, and 14 are also disclosed.

Although various illustrative embodiments are described above, any of a number of changes may be made to various embodiments without departing from the scope of the invention as described by the claims. For example, the order in which various described method steps are performed may often be changed in alternative embodiments, and in other alternative embodiments one or more method steps may be skipped altogether. Optional features of various device and system embodiments may be included in some embodiments and not in others. Therefore, the foregoing description is provided primarily for exemplary purposes and should not be interpreted to limit the scope of the invention as it is set forth in the claims.

The examples and illustrations included herein show, by way of illustration and not of limitation, specific embodiments in which the subject matter may be practiced. As mentioned, other embodiments may be utilized and derived there from, such that structural and logical substitutions and changes may be made without departing from the scope of this disclosure. Such embodiments of the inventive subject matter may be referred to herein individually or collectively by the term “invention” merely for convenience and without intending to voluntarily limit the scope of this application to any single invention or inventive concept, if more than one is, in fact, disclosed. Thus, although specific embodiments have been illustrated and described herein, any arrangement calculated to achieve the same purpose may be substituted for the specific embodiments shown. This disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, will be apparent to those of skill in the art upon reviewing the above description.

While preferred embodiments of the present invention have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions will now occur to those skilled in the art

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without departing from the invention. It should be understood that various alternatives to the embodiments of the invention described herein may be employed in practicing the invention. It is intended that the following claims define the scope of the invention and that methods and structures within the scope of these claims and their equivalents be covered thereby.

What is claimed is:

1. A medical device, comprising:

an elongate body having a first configuration and a second configuration, the second configuration having a different shape from the first configuration;

a plurality of openings on the elongate body configured to allow the elongate body to change shape from the first configuration to the second configuration;

a shuttle axially moveable within the elongate body;

a tension member extending distally from the shuttle and connecting the shuttle to a distal portion of the elongate body;

a pull member extending proximally from the shuttle, wherein pulling the shuttle proximally along the elongate body with the pull member causes the elongate body to transition from the first configuration to the second configuration; and

a locking mechanism configured to hold the shuttle in a proximally pulled position to lock the elongate body in the second configuration.

2. The medical device of claim 1, wherein the elongate body is flexible in the first configuration and rigid in the second configuration.

3. The medical device of claim 1, wherein the elongate body is configured to be connected to an end of a catheter.

4. The medical device of claim 1, wherein the plurality of openings extend from a spine portion of the elongate body.

5. The medical device of claim 4, wherein the spine portion curves around at least a portion of the elongate body.

6. The medical device of claim 4, wherein a width of the openings changes between the first configuration and the second configuration.

7. The medical device of claim 1, wherein the locking mechanism comprises a pin attached to the shuttle that is positioned in and configured to slide along a track extending along a length of the elongate body.

8. The medical device of claim 1, wherein the elongate body further comprises a release window configured to allow disengagement of the pull member from the shuttle.

9. The medical device of claim 1, wherein the locking mechanism is disposed within the elongate body.

10. The medical device of claim 1, wherein the tension member is configured to elongate when the shuttle moves proximally away from the distal portion.

11. A method of advancing a medical device towards a target area, comprising:

advancing an elongate body towards the target area and out of a distal end of a delivery catheter;

manipulating a pull wire to move a shuttle axially within the elongate body to a first position to lock the elongate body into a first curved shape;

manipulating the pull wire to move the shuttle axially into a second position to lock the elongate body into a second curved shape at or near the target area, the second curved shape different from the first curved shape.

12. The method of claim 11, wherein the second position is different from the first position.

13. The method of claim 11, wherein the first curved shape comprises a curve along a first portion of the elongate

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body, and wherein the second curved shape comprises a curve over a second portion of the elongate body, the second portion different from the first portion.

14. The method of claim **13**, wherein the second portion is larger than and encompasses the first portion.

15. The method of claim **13**, wherein the first curved shape comprises a curve at a tip of the elongate body.

16. The method of claim **11**, wherein the target area comprises a native valve in a heart.

17. The method of claim **16**, further comprising advancing the elongate body from a first side of the native valve to a second side of the native valve.

18. The method of claim **16**, further comprising, after moving the medical device into the second curved shape, expanding at least a portion of an expandable valve prosthesis adjacent the native valve.

19. The method of claim **11**, wherein the elongate body comprises a plurality of openings extending from a spine portion of a wall of the device.

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20. The method of claim **11**, wherein the first curved shape is defined by a length of a tension member connecting the shuttle to a distal portion of the elongate member.

21. The method of claim **20**, wherein the length of the tension member is longer when the elongate member assumes the second curved shape.

22. A method of delivering a valve prosthesis anchor to a native valve, comprising:

advancing an elongate body in a flexible delivery configuration towards the native valve and out of a distal end of a delivery catheter;

manipulating a pull wire to move a shuttle axially along the elongate body to cause the elongate body to transition from the flexible delivery configuration to a rigid helical configuration such that the elongate body forms at least one coil around chordae of the native valve; and locking a position of the shuttle within the elongate body to lock the elongate body in the rigid helical configuration around the chordae.

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